

Project: ChainSync – Intelligent Supply Chain Coordination System

1. Introduction

This document outlines the Low-Level Design (LLD) for **ChainSync**, an intelligent supply chain coordination system designed for manufacturers, distributors, and logistics providers. The system digitizes order management, automates inventory tracking, optimizes supplier coordination, and provides analytics for demand forecasting and delivery performance.

Supports backend development using **Java (Spring Boot)** and **.NET (ASP.NET Core)**.

2. Module Overview

- 2.1 Supplier & Product Catalog Management Module
- 2.2 Order Placement & Fulfillment Module
- 2.3 Inventory Tracking & Replenishment Module
- 2.4 Shipment Scheduling & Delivery Tracking Module
- 2.5 Demand Forecasting & Supply Chain Analytics Module

3. Architecture Overview

3.1 Architectural Style

- **Frontend:** Angular/React for supplier, warehouse, and admin portals
- **Backend:** REST APIs for order processing, inventory updates, and analytics
- **Database:** SQL Server/MySQL for structured supply chain data

3.2 Component Interaction

- Frontend captures orders, inventory levels, and delivery updates
- Backend processes fulfillment workflows and generates forecasts

4. Module-Wise Design

4.1 Supplier & Product Catalog Management Module

4.1.1 Features

- Register suppliers and manage product catalogs
- Define lead times, pricing, and availability

4.1.2 Entities

- **Supplier**
 - SupplierID

- Name
- ContactDetails
- ProductList
- LeadTime
- **Product**
 - ProductID
 - Name
 - Category
 - SupplierID
 - UnitPrice
 - AvailabilityStatus

4.2 Order Placement & Fulfillment Module

4.2.1 Features

- Place purchase orders and track fulfillment status
- Manage partial shipments and backorders

4.2.2 Entities

- **Order**
 - OrderID
 - ProductID
 - Quantity
 - SupplierID
 - OrderDate
 - FulfillmentStatus (PENDING, PARTIAL, COMPLETED)

4.3 Inventory Tracking & Replenishment Module

4.3.1 Features

- Monitor stock levels across warehouses
- Trigger automatic replenishment based on thresholds

4.3.2 Entities

- **InventoryItem**
 - ItemID
 - ProductID
 - WarehouseID
 - QuantityAvailable
 - ReorderLevel
 - LastUpdated

4.4 Shipment Scheduling & Delivery Tracking Module

4.4.1 Features

- Schedule shipments and assign carriers
- Track delivery status and estimated arrival

4.4.2 Entities

- **Shipment**
 - ShipmentID
 - OrderID
 - CarrierName
 - DispatchDate
 - EstimatedArrival
 - DeliveryStatus (IN_TRANSIT, DELIVERED, DELAYED)

4.5 Demand Forecasting & Supply Chain Analytics Module

4.5.1 Features

- Analyze historical order data for demand trends
- Generate reports for supplier performance and delivery efficiency

4.5.2 Entities

- **ForecastReport**
 - ReportID
 - ProductID
 - ForecastPeriod
 - PredictedDemand
 - AccuracyScore
- **PerformanceReport**
 - ReportID
 - SupplierID
 - DeliveryTimeliness
 - OrderAccuracy
 - GeneratedDate

5. Deployment Strategy

5.1 Local Deployment

- Developer machines with sample supplier and order data

5.2 Staging and Production Environments

- Cloud deployment with secure APIs and real-time dashboards

6. Database Design

6.1 Tables and Relationships

- Supplier → Product → Order → InventoryItem → Shipment → ForecastReport / PerformanceReport

7. User Interface Design

7.1 Wireframes

- **Supplier Portal:** Manage products, view orders
- **Warehouse Dashboard:** Monitor inventory, schedule shipments
- **Analytics Console:** View forecasts and performance metrics

8. Non-Functional Requirements

8.1 Performance

- Handle 10,000 order transactions per day

8.2 Usability

- Role-based dashboards with intuitive navigation
- Inline help and tooltips for order and inventory workflows

8.3 Security

- Encrypted API communication and secure login
- Compliance with ISO 28000 and data protection standards

8.4 Scalability

- Support for multi-region supply chains and multiple warehouses

9. Assumptions and Constraints

9.1 Assumptions

- Suppliers will update product availability via portal or API
- Warehouses will sync inventory data in real-time

9.2 Constraints

- Initial rollout for domestic supply chain operations only